MAJOR PROJECT · 7th SEMESTER

Satellite-Based Early Warning for Glacial Lake Outburst Floods

Multi-Sensor Anomaly Detection over the Indian Himalaya

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MOTIVATION

Sikkim, 4 October 2023 — the warning was visible for years

55+	50 M	1,200 MW	7 years
lives lost, with 74 more never found	tonnes of water released in a few hours	Teesta-III power station destroyed	the precursor was visible in free satellite data

- A moraine — a wall of loose rock, mud and buried ice left by a retreating glacier — dams these lakes. Nobody engineered it, and nobody inspects it.
- The slope that collapsed into South Lhonak had been creeping measurably for years, and the failure mode had been modelled in a peer-reviewed paper in 2021.
- The signal existed in free, public satellite data the entire time. What was missing was a system that was looking at it.

Everything published so far either counts these lakes, or explains a disaster afterwards. Nothing watches them continuously. That is the gap this project fills.

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Problem Statement

Design and build a decision-support system that continuously monitors Himalayan moraine-dammed glacial lakes using freely available multi-sensor satellite data, learns each lake's normal behaviour without labelled disaster examples, and produces a ranked watch-list of abnormally behaving lakes with the evidence an officer needs to act.

Breaking it down

- Hundreds of moraine-dammed lakes across the Indian Himalaya are unmonitored, and both their number and volume are rising under climate change.
- Precursors — lake growth, moraine slope movement, glacier acceleration, melt intensification — are physically observable weeks to months before failure.
- Sensors cannot be installed: these lakes sit above 4,500 m, days from the nearest road. Satellites already visit every one of them, every few days, free.
- Monsoon cloud removes roughly 35% of optical acquisitions — an optical-only system goes blind exactly in the highest-risk season.
- Only one Indian GLOF is densely observed, so supervised classification is statistically impossible.

Objectives

#	Objective
O1	Build a reproducible ingestion pipeline harmonising optical, radar, ice-velocity, thermal and elevation data for 15–25 Indian Himalayan lakes onto one analysis-ready cube — 10 m grid, 10-day time step.
O2	Integrate Sentinel-1 SAR as an all-weather channel and measure how much observational coverage it recovers during the monsoon.
O3	Develop a downscaling method that makes 1 km MODIS thermal data usable at lake scale.
O4	Develop one-class and self-supervised anomaly models that learn each lake's normal envelope without labelled disasters, benchmarked against a z-score baseline.
O5	Evaluate retrospectively on the 2023 South Lhonak event under a strictly causal protocol, and on lakes never seen in training.
O6	Implement an operational dashboard: ranked watch-list, supporting evidence, and an explicit abstention state.
O7	Publish a multi-sensor precursor benchmark and sensor-ablation study for the Indian Himalaya.

Action verbs: Design · Develop · Implement · Integrate · Analyse · Evaluate

Proposed System Architecture

L1 Data Acquisition	Sentinel-2 L2A · Sentinel-1 GRD/RTC · ITS_LIVE velocity · MODIS LST · Copernicus DEM, pulled from public STAC archives	**Raw scene cache**
L2 Preprocessing	Cloud & QA masking, terrain correction, per-lake UTM reprojection, co-registration to 10 m, 10-day resampling, explicit gap recording	**Harmonised data cube**
L3 Feature Engineering	NDWI lake area, SAR water extent, velocity trend & acceleration, thermal melt index, moraine backscatter change, DEM slope; seasonality removed	**Feature table + observability log**
L4 Modelling & Scoring	Per-lake z-score, Isolation Forest, One-Class SVM, temporal autoencoder; multi-signal fusion; observability-driven abstention	**Risk scores per lake-window**
L5 Decision Support	Ranked watch-list map, per-lake signal explorer with gaps shaded, configurable thresholds, one-click evidence PDF	**Dashboard + evidence report**

One direction of flow. Each layer hands the next a versioned artefact written to disk — never shared memory — so every reported number is reproducible and four students work in parallel without blocking.

Key Methodological Choices

Anomaly detection, not classification

With one confirmed event a classifier would memorise South Lhonak. We model each lake's own normal envelope from years of its history and score deviation. The single event is a held-out test, never a training label.

Multi-signal agreement, not single alarms

Any one channel is noisy. Confidence comes from several independent signals drifting together, which also suppresses false alarms.

Abstention is a first-class output

When cloud or data gaps make a lake unobservable, the system reports "cannot assess" — never a falsely reassuring low score, and never the same colour on screen.

Evidence, not verdicts

The officer receives the score with the underlying time series and before/after imagery. The machine supports a human decision; it never raises a public alarm.

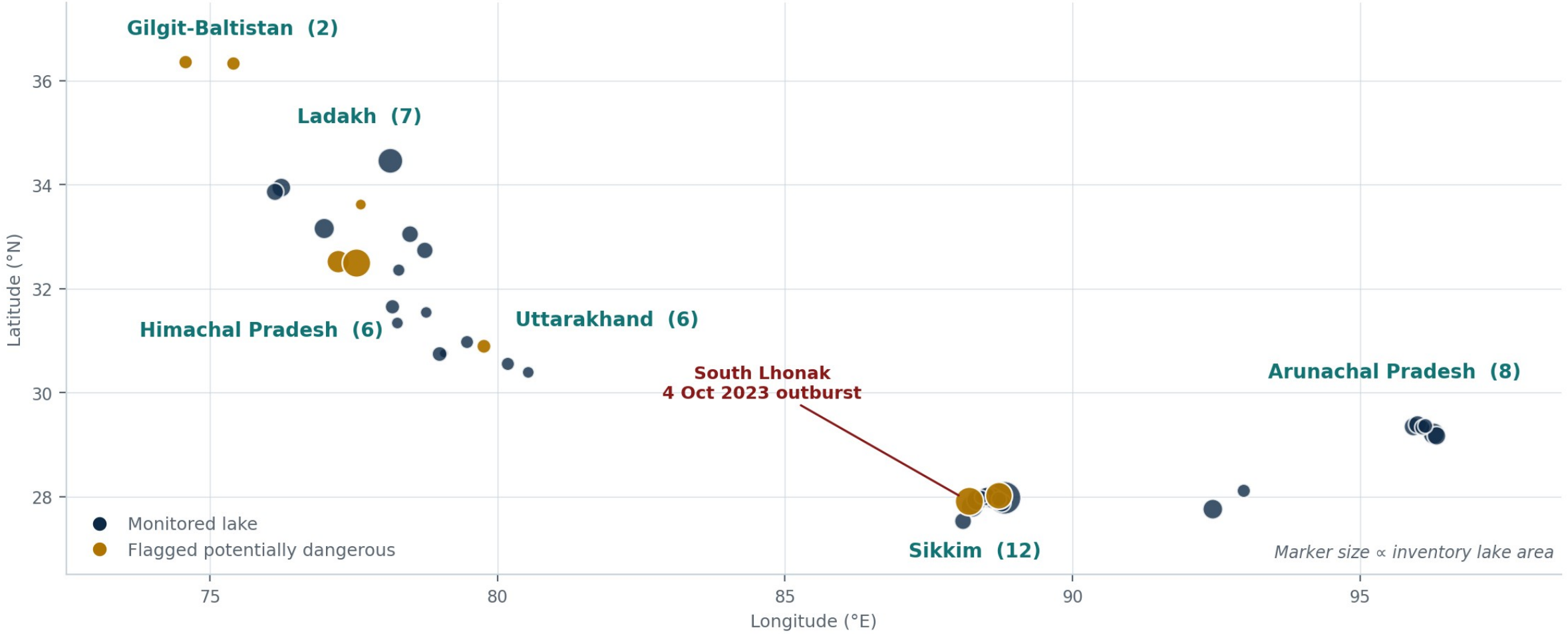
Baseline first: a simple z-score detector is built before any sophisticated model, so every later result is compared honestly.

Team Roles — who owns what

Student	Role	Layer	Responsible for	Hands over
A	Data Collection	L1 – L2	Fetching all satellite data, cloud removal, terrain correction, aligning every sensor onto one grid, and adding the radar channel. The supply line for the whole team.	One tidy data cube that B and C can open in a single line of code
B	Model Development	L3 – L4	Turning raw data into physically meaningful features, then building the models that learn each lake's normal behaviour and score how far it has drifted.	A risk score for every lake, for every 10-day window
C	Testing & Evaluation	Evaluation	Independently checking whether the system works. Writes the test rules before seeing any result, and runs the South Lhonak retrospective.	Every number and chart that appears in the report
D	App Development & Writing	L5	Building the dashboard the officer uses, plus the literature survey, documentation, project report and research paper.	The working app, the report and the paper

Two deliberate design decisions. The split is by function, not by topic, so all four of us are fully occupied from week one and nobody waits for anybody else during the first month. And Student C builds no part of the model — the people who build the system are not the people who judge whether it works, for the same reason an exam is not marked by the candidate.

Study Region — 41 lakes across the Indian Himalaya



Register generated by rule from the NRSC/ISRO Glacial Lake Atlases — 12,868 inventory lakes parsed, 41 selected. 39 sit on inventory polygons; 2 Karakoram lakes use published coordinates. Eight are flagged potentially dangerous.

Dataset Details — Satellite Sources

Dataset	Provider	Resolution	Revisit	Period	Role in the system
Sentinel-2 L2A	ESA Copernicus	10 m	5 days	2017–2024	Lake area and shoreline via NDWI
Sentinel-1 GRD / RTC	ESA Copernicus	10 m	6–12 days	2017–2024	All-weather water extent and moraine backscatter change
ITS_LIVE velocity	NASA JPL	120 m	Annual + seasonal	1985–2024	Glacier velocity trend and acceleration
MODIS MOD11A1 LST	NASA	1 km	Daily	2017–2024	Melt-season thermal index (downscaled)
Copernicus DEM GLO-30	ESA	30 m	Static	—	Slope and moraine-stability proxy

Reference and label datasets

Dataset	What it provides	Use
GLOFNet (2025)	Multimodal Sentinel-2 + ITS_LIVE + MODIS patches	Primary published dataset; lets modelling start in week one
HMA GLOF Database (2023)	697 documented outburst floods, 1833–2022	Truth list for retrospective evaluation
NRSC/ISRO Glacial Lake Atlases	12,868 inventory lakes across Ganga, Brahmaputra, Indus and IHR basins	Rule-based selection of the study register
ICIMOD inventories	Himalayan lake inventory with expert danger ratings	Cross-check on lake selection

Every dataset is public and free. Total data cost: ₹0. Study register: 41 lakes across Sikkim, Uttarakhand, Himachal Pradesh, Ladakh, Arunachal Pradesh and Gilgit-Baltistan; 10-day windows, 2017–2024.

Literature Review — What exists, and what does not

What researchers have done	Representative work	What it does not do
Lake inventories and growth studies	Shugar et al., Nature Climate Change 2020 · Kumar & Vijay, Scientific Reports 2026	A headcount at one moment. Tells you a lake exists, not how it is behaving today.
Post-event forensic reconstruction	Sattar et al., Science 2025 (Sikkim cascade)	Excellent science that arrives after the flood. It cannot warn anyone.
Hazard and dam-break modelling	Sattar et al., Geomorphology 2021 (South Lhonak)	Answers "what if it bursts", not "is it about to burst". No timing of failure.
Event databases	Shrestha et al., ESSD 2023 — 697 GLOFs, 1833–2022	Supplies labels, and honestly reports that most historical events were never recorded.
Multimodal ML for GLOF	GLOFNet 2025 · IceWatch 2026 (arXiv)	Optical-only, Karakoram-only, and neither can declare a lake unobservable.

The gap is stated by the prior authors themselves. The GLOFNet paper names four limitations — extreme class imbalance, cloud contamination, cross-sensor resolution mismatch and velocity-tracking artefacts — and lists its own future directions as adding SAR, improving thermal downscaling and expanding to more basins. Those are objectives O2, O3 and O1 of this project.

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SDG Mapping

SDG 13.1 Climate Action	Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters — the system provides advance situational awareness for a hazard driven directly by glacier retreat.
SDG 11.5 Sustainable Cities & Communities	Reduce deaths and people affected by disasters, and decrease direct economic losses — the 2023 Sikkim event killed at least 55 people and destroyed a 1,200 MW power station.
SDG 6.6 Clean Water & Sanitation	Protect and restore water-related ecosystems, including mountains — continuous monitoring of high-altitude lakes supports mountain water-system stewardship.

Beyond the metrics: the system costs nothing to run. Every dataset is public and every compute platform is a free tier, so a resource-constrained state agency could adopt it without a procurement budget.

Expected Outcomes

Deliverables

- An end-to-end pipeline monitoring the study lakes from free archives, rebuildable from a single documented command.
- A Streamlit decision-support dashboard — ranked watch-list, per-lake signal explorer with gaps visible, configurable thresholds, downloadable evidence PDF.
- A versioned harmonised multi-sensor data cube and feature store, reusable by future batches.
- A workshop paper, project report, SRS, design document and a public reproducible repository.

How success is measured

Metric	What it establishes
Lead time	How far ahead of 4 Oct 2023 the system would have flagged South Lhonak, using only pre-event data
Observability recovery (%)	Extra monitored windows gained from Sentinel-1 during the monsoon versus optical alone
False alarms per lake-year	Whether the alarm burden is one an officer could actually absorb
Abstention rate (%)	How often the system honestly declares a lake unobservable, and when
Ablation deltas	Which sensors a real monitoring programme would actually need to pay for
Generalisation gap	Leave-one-lake-out versus in-sample — whether the model learned or memorised

Insurance against a null result: even a partly negative retrospective still delivers the first multi-sensor benchmark for the Indian Himalaya and a measured account of why precursor detection is hard under single-event supervision.

The warning signs were visible in free satellite data for seven years before Sikkim.

The information existed. No system was watching it.

We are building the system that watches.

Thank you